



ASSIGNMENT

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- 1) What If There Were No Operating System? Imagine a computer without an operating system. Identify the problems a user would face and investigate how OS features solve them. Demonstrate at least three features using Linux and present a “*Without OS vs with OS*” comparison.

Answer:

As we already know, an operating system (OS) is a system software that manages a computer’s hardware and software resources and provides common services for computer programs. In reality, the earliest computers didn't have operating systems; they were huge machines tasked with one program at a time.

Without an operating system, a computer can still function, but every task must be programmed manually. You would need to write code for everything—from recognizing keyboard inputs to displaying characters and handling every feature of an application. An operating system simplifies this by managing the computer hardware and providing a standard platform for applications to run.

Problems a user would face without an OS are:

- No User Interface
- No Application Execution
- Boot Failure
- No Resource Management
- Limited Functionality

THREE LINUX FEATURES DEMONSTRATED

- **File System Management** – *ls* lists files; *mkdir* creates directories.
- **Process Management** – *ps* shows running processes; *kill* terminates them.
- **Device Access** – *dmesg* reads kernel messages; *mount* attaches storage devices.

Example:

```
~ ~ ~bash
$ mkdir test.txt
$ ls
$ ps
$ dmesg | tail
~ ~ ~
```

Without OS v/s With OS (Linux) Comparison

Feature	Without OS	With OS
Boot	BIOS/UEFI may stop at “No bootable device”	Linux kernel loads and mounts the root of file system.
User Interface	No standard UI	Desktop environments like GNOME/KDE or CLI are available.
File Access	No OS-managed filesystem	Commands like <i>ls</i> , <i>cd</i> , and <i>cp</i> work
Process Execution	No standard process management	Shells like bash and applications can run
Resource Management	No OS to manage CPU, memory, or I/O	Linux kernel manages CPU, memory and I/O resources

- 2) Imagine your team is setting up a cybersecurity laboratory. Explore at least five Linux distributions relevant to cybersecurity and recommend suitable distributions for different roles such as penetration testing, digital forensics,

privacy, and security monitoring. Present your findings creatively as a comparison chart, poster, infographic, or “OS selection guide.”

Answer:

LINUX FOR CYBERSECURITY

EXPLORE. ANALYZE. SECURE.

OS SELECTION GUIDE FOR YOUR CYBERSECURITY LAB

DISTRIBUTION >	KALI LINUX	PARROT OS SECURITY	CAINE	Tails	SECURITY UNION
	The most advanced penetration testing distribution.	Security, privacy and development in one OS.	Computer Aided Investigative Environment.	The Amnesic Incognito Live System focused on privacy.	Linux distro for security monitoring and log analysis.
PRIMARY FOCUS	Penetration Testing & Ethical Hacking	Privacy, Pentesting, Development	Digital Forensics & Incident Response	Privacy, Anonymity & Secure Browsing	Security Monitoring & Threat Detection
BASED ON	Debian	Debian	Ubuntu LTS	Debian (Stable)	Debian
PACKAGE MANAGER	apt	apt	apt	apt	apt
PRE-INSTALLED TOOLS (EXAMPLES)	- Metasploit - Nmap - Burp Suite - Wireshark - SQLmap - Aircrack-ng	- Metasploit - Wireshark - John the Ripper - Burp Suite - Ghidra - Anonsurf	- Autopsy - FTK Imager - Sleuth Kit - Volatility - Wireshark - PhotoRec	- Tor Browser - KeePassX - Electrum - GnuPG - Onionshare - ZuluCrypt	- Zeek (Bro) - Suricata - Wazuh - Elastic Stack - Wireshark - OSQuery
BOOT TYPE	Live or Persistent Live	Live or Persistent Live	Live (Forensics Bootable)	Live (Read-only)	Live or Installation
BEST FOR	Penetration Testers, Red Teamers, Security Researchers	Security Enthusiasts, Developers, Privacy Advocates	Forensic Analysts, Incident Responders, Law Enforcement	Journalists, Activists, Whistleblowers, Privacy Seekers	SOC Analysts, Blue Team, Network Admins
STRENGTHS	Huge toolset, community support, rolling updates, optimized for hacking.	Lightweight, privacy focused, good for development & cloud.	Forensic ready, excellent hardware support, easy to use.	Routes all traffic through Tor, leaves no trace.	Powerful monitoring stack, excellent visualization.
LIMITATIONS	Not ideal for daily use, can be heavy for beginners.	Some tools need manual configuration for best results.	Not for general usage, specialized for forensics only.	Limited persistence, not for heavy computing tasks.	Requires more resources for full installation.
OFFICIAL SITE	https://www.kali.org	https://www.parrotsec.org	https://www.caine-live.net	https://tails.net	https://securityonion.net

QUICK TIPS

Always run Linux distros in a VM or isolated environment.

Keep your OS & tools updated regularly.

Use strong passwords & encryption.

Follow legal and ethical hacking practices.

THE RIGHT OS CAN EMPOWER YOUR SKILLS. CHOOSE WISELY. STAY ETHICAL. STAY SECURE.

- 3) Exit status: Execute at least five Linux commands that produce both successful and unsuccessful results. Investigate their exit status using `$?`. Create a simple visual/table showing *Command* → *Result* *Exit Status* → *Meaning*. What can a script learn from an exit status?

[ANSWER UPLOADED AS AN .md FILE IN GIT REPOSITORY]